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**DATA MINING FINAL PROJECT REPORT**

**Healthcare Patient Analysis using Clustering & Classification**

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## 1. Business Understanding

### 1.1 Problem Definition

Hospitals face a major challenge in managing patient readmissions, which significantly increases healthcare costs and reduces the quality of care. Understanding which types of patients are more likely to be readmitted allows healthcare providers to improve treatment strategies, allocate resources efficiently, and reduce unnecessary hospital returns.

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### 1.2 Objective

The main objective of this project is to apply data mining techniques to:

- Segment patients into meaningful groups using clustering
- Predict patient readmission using classification models
- Extract actionable insights that can support healthcare decision-making

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### 1.3 Business Questions

This project aims to answer the following questions:

- Can patients be grouped into meaningful clusters based on their medical and demographic data?
- Which factors contribute most to patient readmission?
- Can we predict whether a patient will be readmitted?
- What actionable insights can hospitals use to improve patient care?

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### 1.4 Success Criteria

The project is considered successful if:

- Meaningful patient clusters are identified
- A classification model achieves reasonable predictive performance
- Key features influencing readmission are identified
- A working interactive dashboard is developed for exploration

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## 2. Data Understanding

### 2.1 Dataset Description

The dataset used in this project contains hospital patient records including demographic, clinical, and administrative information.

**Key Features:**

- age
- gender
- primary_diagnosis
- num_procedures
- days_in_hospital
- comorbidity_score
- discharge_to
- readmitted (target variable)

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### 2.2 Dataset Size

- Number of records: depends on provided dataset
- Number of features: multiple numeric and categorical attributes

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### 2.3 Data Quality Issues

During exploration, the following issues were identified:

- Missing values in numerical columns
- Categorical variables requiring encoding
- Mixed data types
- Need for scaling numerical features

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### 2.4 Exploratory Data Analysis (EDA)

Initial analysis included:

- Distribution of numeric variables
- Frequency of categorical features
- Correlation between features
- Readmission rate distribution

Key observation:
Some features such as age, comorbidity score, and hospital stay duration show potential influence on readmission patterns.

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## 3. Data Preparation

### 3.1 Data Cleaning

To ensure high-quality input for modeling:

- Missing numerical values were filled using median imputation
- Categorical missing values were filled with "Unknown"
- Data types were standardized

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### 3.2 Feature Encoding

Categorical variables were converted into numerical format using:

- One-hot encoding (pd.get_dummies)

This ensured compatibility with machine learning models.

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### 3.3 Feature Scaling

To ensure fair contribution of all features:

- StandardScaler was applied to normalize numerical values

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### 3.4 Feature Selection

Two sets of features were used:

**Numerical features:**

- age
- num_procedures
- days_in_hospital
- comorbidity_score

**Categorical features:**

- gender
- primary_diagnosis
- discharge_to

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### 3.5 Data Split

For classification tasks:

- Training set: 80%
- Validation set: 20%

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## 4. Modeling

### 4.1 Clustering Model

**Algorithm Used: K-Means Clustering**

K-Means was selected to group patients into homogeneous clusters based on similarity.

**Steps:**

- Data preprocessing (encoding + scaling)
- Elbow method used to determine optimal clusters
- K = 3 clusters selected
- PCA applied for 2D visualization

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### 4.2 Clustering Output

The clustering model produced:

- Distinct patient groups
- Cluster labels assigned to each patient
- PCA visualization for interpretability

**Evaluation Metric:**

- Silhouette Score used to evaluate cluster quality

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### 4.3 Classification Model

**Algorithm Used: Random Forest Classifier**

Random Forest was chosen due to:

- Ability to handle mixed data types
- Strong performance on structured data
- Feature importance interpretability

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**Steps:**

- Data encoding and scaling
- Train-test split (80/20)
- Model training
- Prediction on validation set

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### 4.4 Classification Output

The model produced:

- Predictions for readmission
- Probability-based decision structure
- Feature importance ranking

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### 4.5 Feature Importance

The most influential factors in predicting readmission were:

- Length of hospital stay
- Comorbidity score
- Age
- Primary diagnosis

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## 5. Evaluation

### 5.1 Clustering Evaluation

- Method: Silhouette Score
- Purpose: Measure how well-separated clusters are

Result:

- Moderate clustering separation indicating meaningful but overlapping patient groups

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### 5.2 Classification Evaluation

**Metrics Used:**

- Accuracy
- Precision

- Recall
- F1-score
- Confusion Matrix

**Interpretation:**

- The model provides reasonable predictive ability
- Some misclassifications exist due to overlapping patient characteristics

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### 5.3 Business Interpretation

- Patients with longer hospital stays and higher comorbidity scores are more likely to be readmitted
- Certain diagnosis categories show higher readmission rates
- Clustering helps identify high-risk patient segments

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## 6. Deployment (Streamlit Dashboard)

### 6.1 Overview

A Streamlit-based interactive dashboard was developed to deploy the data mining results.

The dashboard allows users to:

- Upload dataset
- Switch between clustering and classification modes
- Visualize patient clusters
- View classification performance
- Explore feature importance
- Generate business insights

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### 6.2 Dashboard Features

**Clustering Section:**

- K-Means clustering results
- PCA visualization
- Cluster size distribution
- Cluster profiling (numeric & categorical)

**Classification Section:**

- Model accuracy
- Confusion matrix
- Classification report
- Feature importance visualization

**Business Insights Panel:**

Automatically generated insights such as:

- High-risk patient clusters
- Key influencing medical factors
- Resource allocation recommendations

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### 6.3 Value of Deployment

The dashboard transforms raw data into:

- Interactive decision support tool
- Visual analytics system
- Business intelligence interface

This aligns with real-world healthcare analytics systems.

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## 7. Conclusion

This project successfully demonstrates the full CRISP-DM pipeline applied to a healthcare dataset.

**Key Achievements:**

- Patient segmentation using clustering
- Readmission prediction using classification
- Feature importance analysis
- Interactive dashboard deployment

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**Final Insight:**

Data mining techniques can significantly support healthcare decision-making by identifying high-risk patients and improving resource allocation strategies.

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**8. Future Work**

Possible improvements include:

- Testing advanced clustering methods (DBSCAN, GMM)
- Using deep learning models for prediction
- Adding temporal patient data
- Improving dashboard with real-time updates
- Deploying online via Streamlit Cloud or AWS